Time and Space Complexity

A. Natural Sum

0.5 seconds, 256 megabytes

You are given a single integer N. Your task is to compute the sum of the first N natural numbers.

Input

- A single integer N $(1 \le N \le 2 \cdot 10^9)$.

Output

Output a single integer — the sum of the first N natural numbers.

input
```
5
```
output
```
15
```

input
```
10
```
output
```
55
```

The sum of the first 5 natural numbers is $1 + 2 + 3 + 4 + 5 = 15$.

B. Interval Sum

1 second, 256 megabytes

You are given Q queries. Each query consists of two integers L and R ($1 \le L, R \le 10^9$). For each query, compute the value:

L + (L+1) + (L+2) + ... + (R-1) + R

In other words, find the sum of all integers from L to R (inclusive). It is guaranteed that $L \le R$.

Input

- The first line contains a single integer Q $(1 \le Q \le 10^5)$.
- The next Q lines each contain two integers L and R.

Output

For each query, output the sum from L to R on a new line.

input
```
3
1 3
5 5
2 6
```
output
```
6
5
20
```

C. Counting Intervals

1 second, 256 megabytes

You are given Q queries. Each query consists of three integers T, L, R.

For each query, you must count how many integers lie in the range defined by L and R depending on T:

- $T = 1$: Count integers in the interval (L, R) (exclusive on both ends)
- $T = 2$: Count integers in the interval $[L, R)$ (inclusive on left, exclusive on right)

- $T = 3$: Count integers in the interval $(L, R]$ (exclusive on left, inclusive on right)
- $T = 4$: Count integers in the interval $[L, R]$ (inclusive on both ends)

Input

- Integer Q $(1 \le Q \le 10^5)$
- Each of the next Q lines contains three integers: T, L, R. ($-10^{18} \le L, R \le 10^{18}$, $1 \le T \le 4$).

Output

For each query, output the required count. If $L > R$ output 0.

input

```
4
1 2 7
2 2 7
3 2 7
4 2 7
```

output

```
4
5
5
6
```

For $L = 2, R = 7$:

$(2, 7) = \{3, 4, 5, 6\}$ has 4 integers.

$[2, 7) = \{2, 3, 4, 5, 6\}$ has 5 integers.

$(2, 7] = \{3, 4, 5, 6, 7\}$ has 5 integers.

$[2, 7] = \{2, 3, 4, 5, 6, 7\}$ has 6 integers.

D. Count Factors

1 second, 256 megabytes

Given an integer N, count the number of factors* of N.

* A **factor** (or divisor) of a number N is a positive integer d such that $N \bmod d = 0$.

Input

A single integer N $(1 \le N \le 10^{12})$.

Output

Output a single integer — the total number of positive divisors of N.

input

```
12
```

output

```
6
```

input

```
17
```

output

```
2
```

Example 1: The positive divisors of 12 are: $[1, 2, 3, 4, 6, 12]$. Therefore, the answer is 6.

E. Am I Prime?

1 second, 256 megabytes

You are given a single integer N. Your task is to determine whether N is a prime number.

Input

A single integer N $(1 \le N \le 10^{12})$.

Output

Print `YES` if N is prime, otherwise print `NO`.

input

```
12
```

output

```
NO
```

input

```
17
```

output

```
YES
```

F. Count Primes

1 second, 256 megabytes

Given an integer N, find how many prime numbers are less than or equal to N.

Input

A single integer N $(1 \le N \le 10^5)$.

Output

Output a single integer — the count of primes $\le N$.

input

```
10
```

output

```
4
```

input

```
100
```

output

```
25
```

Example 1: Primes ≤ 10 are: $2, 3, 5, 7$. Thus the answer is 4.

G. Print Factors

1 second, 256 megabytes

Given a positive integer N, print all its positive factors* in **ascending order**.

* A **factor** (or divisor) of a number N is a positive integer d such that $N \bmod d = 0$.

Input

A single integer N $(1 \le N \le 10^{12})$.

Output

Print all positive factors of N in increasing order

input

```
12
```

output

```
1 2 3 4 6 12
```

input

```
103
```

output

```
1 103
```

H. Kth Factor

1 second, 256 megabytes

You are given two integers N and K. Your task is to determine the K-th smallest factor of N.

If N has fewer than K factors, print `-1`.

Input

Two integers N and K $(1 \le N \le 10^{12},\ 1 \le K \le 10^{12})$.

Output

Print the K-th smallest factor of N, or -1 if it does not exist.

input
```
12 3
```
output
```
3
```

input
```
103 3
```
output
```
-1
```